Parametric Hypothesis Tests — Part 1

Basic Concepts, Test Procedure, and Errors

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Introduction to Hypothesis Testing

Recall: Statistical Inference

So far in this course, we have covered:

- **Sampling distributions** (Topic 1): how sample statistics behave
- **Point estimation** (Topic 2.1): finding a single "best guess" for a parameter
- **Interval estimation** (Topic 2.2): constructing a range of plausible values

Now we ask a different question:

> Given a claim about a population parameter, does the sample data support or contradict that claim?

Why Hypothesis Testing?

In practice, decisions must be made under uncertainty:

- A pharmaceutical company claims a new drug lowers cholesterol by more than 20 mg/dL. Does the clinical trial data support this?
- An engineer suspects that a machine is overfilling packages beyond the nominal 100 g. Is there statistical evidence?
- A bank manager believes the average processing time for loan applications has decreased. Can we confirm this?

Hypothesis testing provides a **formal, rigorous framework** for answering these questions using sample data.

From Estimation to Testing

The Key Shift

In **estimation**, we ask: *“What is the value of the parameter?”*

In **hypothesis testing**, we ask: *“Is the parameter equal to (or greater than, or less than) a specific value?”*

We already have the building blocks:

- Sampling distributions
- Pivot (fulcral) variables: Z, T, Q (χ^2)
- Critical values from statistical tables

3.1 — Basic Concepts

The Structure of a Hypothesis Test

Every hypothesis test involves the same fundamental elements:

1. **Null hypothesis** (H_0): the claim we assume to be true until evidence says otherwise
2. **Alternative hypothesis** (H_1): the claim we are trying to find evidence for
3. **Test statistic**: a function of the sample data (computed under H_0)
4. **Significance level** (α): the probability of rejecting H_0 when it is actually true
5. **Critical (rejection) region**: the set of values of the test statistic

The Null Hypothesis (H_0)

Null Hypothesis

The **null hypothesis**, denoted H_0, represents the *status quo* — the current belief, the manufacturer's specification, or the default assumption.

It always contains an **equality** sign ($=$, $\leq$, or $\geq$).

Examples:

- $H_0: \mu = 100$ (the mean weight is 100 g)
- $H_0: \mu \leq 500$ (the mean daily revenue does not exceed 500 €)
- $H_0: \sigma^2 \geq 10$ (the variance is at least 10)

> **Important**: We never "accept" H_0. We either **reject** it or **fail to reject** it.

The Alternative Hypothesis (H_1)

Alternative Hypothesis

The **alternative hypothesis**, denoted H_1 (or H_a), represents what we are trying to find evidence **for**. It is the “research hypothesis.”

It contains a strict inequality ($\neq$, $>$, or $<$).

The form of H_1 determines the **type of test**:

H_0	H_1	Type of test
$\mu = \mu_0$	$\mu \neq \mu_0$	Two-tailed (bilateral)
$\mu \leq \mu_0$	$\mu > \mu_0$	Right-tailed (unilateral right)
$\mu \geq \mu_0$	$\mu < \mu_0$	Left-tailed (unilateral left)

How to Formulate H_0 and H_1

A common source of confusion: *which claim goes in H_0 and which in H_1?*

Rule of thumb:

- H_0 is the claim that we assume true unless the data convinces us otherwise
- H_1 is the claim we want to **prove** (what the researcher suspects)

> Think of it as a **trial**: H_0 is "innocent" (the default), and H_1 is "guilty." The burden of proof lies on H_1.

Example: Formulating Hypotheses

Scenario: A food company states that each package contains, on average, 500 g of product. A consumer protection agency suspects the company is underfilling.

What are H_0 and H_1?

- The *status quo* (company's claim): $\mu = 500$ (or $\mu \geq 500$)
- The *suspicion* (what we want evidence for): $\mu < 500$

$$H_0 : \mu \geq 500 \quad \text{vs.} \quad H_1 : \mu < 500$$

This is a **left-tailed** test.

Example: Formulating Hypotheses (cont.)

Scenario: An engineer suspects that a machine is overfilling packages beyond the nominal 100 g.

$$H_0 : \mu \leq 100 \quad \text{vs.} \quad H_1 : \mu > 100$$

This is a **right-tailed** test.

Scenario: A quality control inspector wants to check whether the mean weight differs from the specification of 100 g (could be above or below).

$$H_0 : \mu = 100 \quad \text{vs.} \quad H_1 : \mu \neq 100$$

The Test Statistic

Test Statistic

The **test statistic** is a random variable, computed from the sample, whose distribution is known **under** H_0.

It measures *how far* the sample result is from what H_0 predicts.

You already know these from interval estimation:

Parameter	Conditions	Test Statistic	Distribution under H_0
μ	σ known	$Z_0 = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$	$N(0,1)$
μ	σ unknown	$T_0 = \frac{\bar{X}-\mu_0}{S'/\sqrt{n}}$	$t_{(n-1)}$
σ^2	Normal pop.	$Q_0 = \frac{(n-1)S'^2}{\sigma_0^2}$	$\chi^2_{(n-1)}$

The Significance Level (α)

Significance Level

The **significance level** α is the maximum probability of rejecting H_0 when H_0 is actually true.

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

Common values: $\alpha = 0.01$, $\alpha = 0.05$, $\alpha = 0.10$.

- Smaller $\alpha \Rightarrow$ harder to reject $H_0 \Rightarrow$ stronger evidence required
- Larger $\alpha \Rightarrow$ easier to reject $H_0 \Rightarrow$ weaker evidence suffices

> α is chosen **before** collecting data and performing the test.

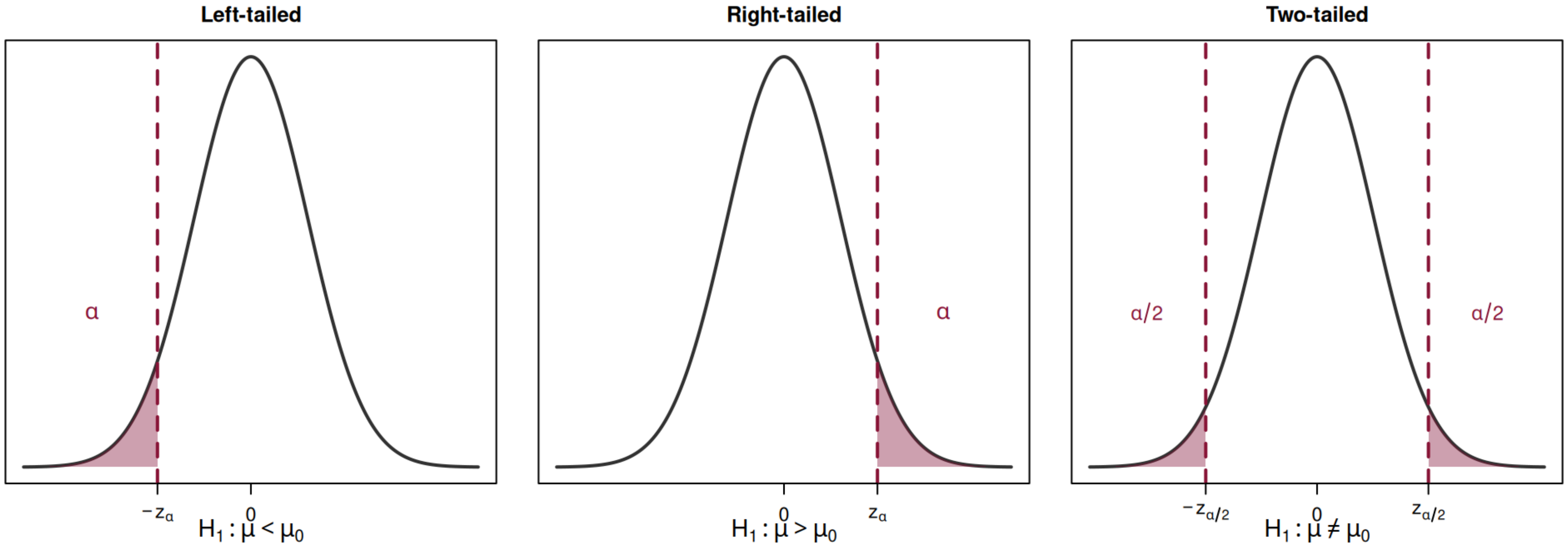
The Critical (Rejection) Region

Critical Region

The **critical region** (or rejection region) is the set of values of the test statistic for which we reject H_0.

Its boundary is determined by the significance level α and the type of test.

Critical Regions: Visual Summary



The shaded areas represent the rejection region. If the observed test statistic falls in the shaded area, we reject H_0.

The General Test Procedure

Step-by-step procedure for a hypothesis test

1. **State the hypotheses**: write H_0 and H_1
2. **Choose the significance level** α
3. **Identify the test statistic** and its distribution under H_0
4. **Determine the critical region** (using α and statistical tables)
5. **Compute the observed value** of the test statistic from the sample
6. **Make the decision**: reject H_0 if the observed value falls in the critical region; otherwise, do not reject H_0
7. **State the conclusion** in the context of the problem

Type I and Type II Errors

Two Types of Mistakes

When we make a decision based on sample data, we can make two kinds of errors:

	H_0 **is true**	H_0 **is false**
Do not reject H_0	Correct decision	**Type II Error** (β)
Reject H_0	**Type I Error** (α)	Correct decision

Type I Error

Type I Error

A **Type I error** occurs when we **reject** H_0 even though H_0 is **true**.

$$\alpha = P(\text{Type I Error}) = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

This is exactly the **significance level** α.

Analogy: Convicting an innocent person in a trial.

Type II Error

Type II Error

A **Type II error** occurs when we **fail to reject** H_0 even though H_0 is **false** (i.e., H_1 is true).

$$\beta = P(\text{Type II Error}) = P(\text{do not reject } H_0 \mid H_1 \text{ is true})$$

Analogy: Acquitting a guilty person in a trial.

β depends on:

- The true value of the parameter (how far from H_0)
- The sample size n
- The significance level α

The Power of a Test

Power

The **power** of a test is the probability of correctly rejecting H_0 when H_1 is true:

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ is true})$$

A good test has **high power** (close to 1).

In practice:

- Increasing n increases power (and reduces β)
- Increasing α increases power but also increases the risk of Type I error
- There is always a **trade-off** between α and β

The Trade-off Between α and β

For a **fixed sample size** n:

- If we decrease α (more conservative) $\Rightarrow$ β increases (harder to detect a real effect)
- If we increase α (more liberal) $\Rightarrow$ β decreases (easier to detect, but more false alarms)

> The only way to reduce **both** α and β simultaneously is to **increase the sample size** n.

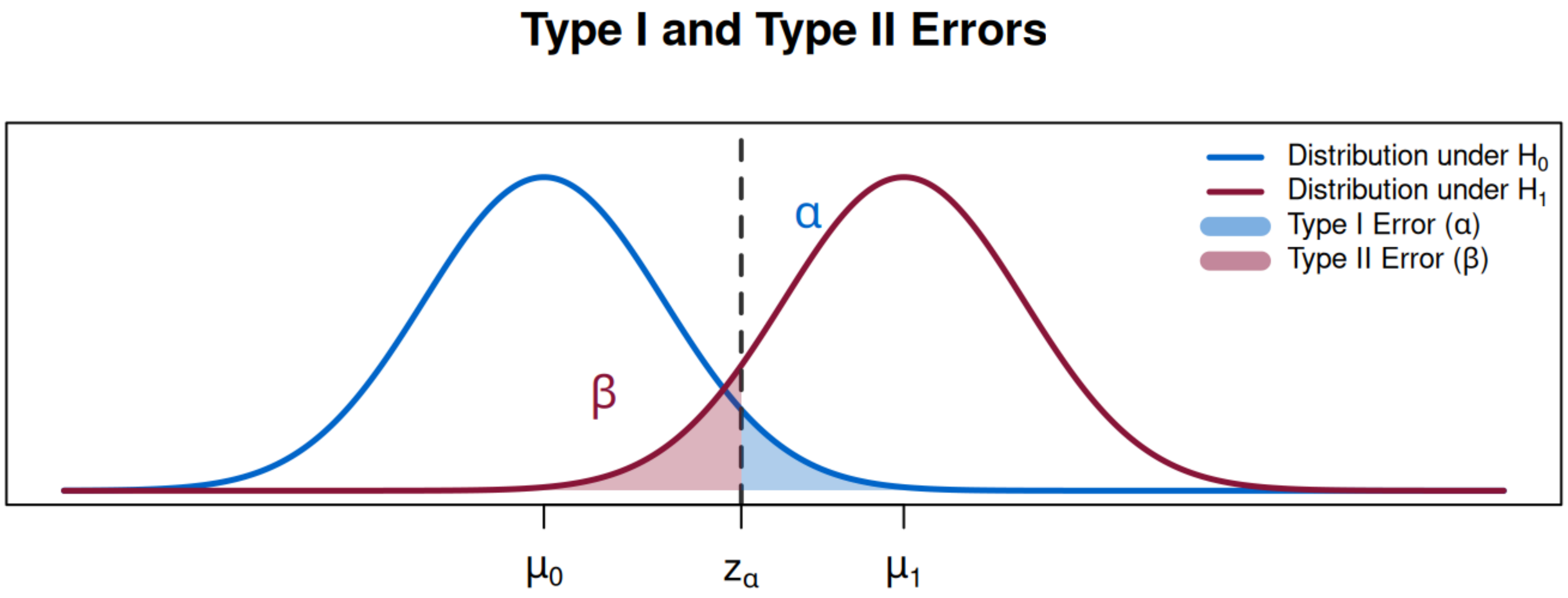
Analogy: Fire Alarm

Think of a hypothesis test as a fire alarm:

- **Type I Error** (α): The alarm goes off, but there is no fire (false alarm)
- **Type II Error** (β): There is a fire, but the alarm does not go off (missed detection)

- If we make the alarm **very sensitive**: fewer missed fires (β small), but more false alarms (α large)
- If we make the alarm **less sensitive**: fewer false alarms (α small), but more missed fires (β large)

Summary: Errors at a Glance



Solved Example — Putting It All Together

Example 1: Test for the Mean (σ known)

A bottling company fills bottles with a nominal volume of $\mu_0 = 500$ mL. The filling process is known to have a standard deviation of $\sigma = 8$ mL. A quality inspector takes a random sample of $n = 36$ bottles and obtains a sample mean of $\bar{x} = 497$ mL.

At the 5% significance level, is there evidence that the machine is underfilling?

Example 1: Solution — Step 1

Step 1: State the hypotheses.

The inspector suspects underfilling, i.e., the mean is *less than* 500.

$$H_0 : \mu \geq 500 \quad \text{vs.} \quad H_1 : \mu < 500$$

This is a **left-tailed** test.

Example 1: Solution — Step 2

Step 2: Significance level.

$$\alpha = 0.05$$

Example 1: Solution — Step 3

Step 3: Test statistic.

The population standard deviation $\sigma = 8$ is **known**, so:

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \underset{\text{under } H_0}{\sim} N(0, 1)$$

Example 1: Solution — Step 4

Step 4: Critical region.

For a left-tailed test at $\alpha = 0.05$:

$$\text{Rejection region: }]-\infty\,;\,-z_\alpha] =]-\infty\,;\,-1.645]$$

We reject H_0 if $z_{obs} \leq -1.645$.

Example 1: Solution — Step 5

Step 5: Compute the observed value.

$$z_{obs} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{497 - 500}{8/\sqrt{36}} = \frac{-3}{8/6} = \frac{-3}{1.333} \approx -2.25$$

Example 1: Solution — Step 6

Step 6: Decision.

$$z_{obs} = -2.25 \leq -1.645$$

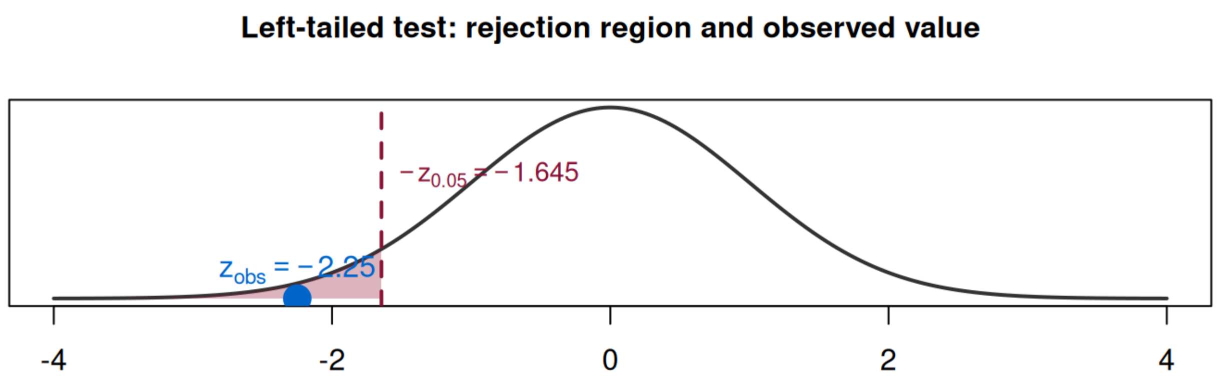
The observed value **falls in the rejection region**.

$\Rightarrow$ We **reject** H_0 at the 5% significance level.

Example 1: Solution — Step 7

Step 7: Conclusion.

At the 5% significance level, there is **sufficient statistical evidence** to conclude that the machine is underfilling the bottles (i.e., the mean volume is less than 500 mL).



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Example 2: Formulating Hypotheses — Practice

For each scenario, write H_0 and H_1, and state the type of test:

a. A bank claims the average time to process a loan is 5 days. A consultant believes it takes longer.

b. A manufacturer states the defect rate is at most 3%. An auditor suspects it is higher.

c. A researcher wants to test if a training program changes employee productivity from the current average of 40 units/day.

Example 2: Solutions

a. $H_0 : \mu \leq 5 \quad \text{vs.} \quad H_1 : \mu > 5$ → Right-tailed

b. $H_0 : p \leq 0.03 \quad \text{vs.} \quad H_1 : p > 0.03$ → Right-tailed

c. $H_0 : \mu = 40 \quad \text{vs.} \quad H_1 : \mu \neq 40$ → Two-tailed

> **Note**: In (c), the researcher does not have a directional suspicion — it could go either way — hence the two-tailed test.

Quick Recap Before the Break

What we covered so far

- **Null hypothesis** (H_0): status quo, contains $=$, $\leq$, or $\geq$
- **Alternative hypothesis** (H_1): research claim, contains $\neq$, $>$, or $<$
- **Test statistic**: pivot variable computed under H_0
- **Significance level** (α): $P(\text{reject } H_0 \mid H_0 \text{ true})$
- **Critical region**: values that lead to rejection of H_0
- **Type I Error** (α): rejecting a true H_0
- **Type II Error** (β): not rejecting a false H_0
- **Power** = $1 - \beta$

After the break: the p**-value** approach (Section 3.2).